MOCK TEST 68

📖 READING

5 parts · 35 questions

⏱ 60 minutes

Vaqt: 60 daqiqa. Javoblarni javoblar varaqasiga o'tkazishni unutmang.

PART 1

Questions 1–6

Read the text. Fill in each gap with ONE word. You must use a word which is somewhere in the rest of the text.

POLAR BEARS

Polar bears are decreasing in number because of climate change. Polar bears depend on Arctic sea **Q1**_____ to hunt for their food. In a warming climate, sea ice is disappearing. Because of this, many polar bears around the Arctic are expected to disappear. Are there any places in the **Q2**_____ where polar bears might have a chance to survive when the **Q3**_____ ice disappears? What would those habitats look like? Would the polar bears that live there look or act differently? We **Q4**_____ a group of polar bears in Southeast Greenland that live in fjords. They use ice from glaciers as well as sea ice to hunt. We conducted a study looking at their movement, genetics, and demography. We found that Southeast Greenland polar bears don't travel much because of the rugged environment. **Q5**_____ bears have been isolated from other polar bears for about 200 years and are a genetically distinct **Q6**_____ . Future studies of these polar bears will help with conservation efforts for the species.

PART 2

Questions 7–13

Look at the five online reviews of the Wellington Hotel, A-E. Which review mentions the following? Write the correct letter, A-E, in boxes 7-13 on your answer sheet. NB You may use any letter more than once.

7
liking the view from the bedroom window

8
finding the receptionists welcoming

9
being pleased with the bedroom

10
becoming confused when booking a room

11
being impressed by the wide choice of food

12
staying in the hotel regularly

13
finding it inconvenient that the hotel was crowded

A) My husband and I first stayed at the Wellington a few years ago, and we've returned every year since then. When we arrive and check in, we're always treated like old friends by the staff, so we very much feel at home. Our one disappointment during our last visit was that our room overlooked the car park, but that didn't spoil our stay.

B) The hotel hardly seems to have changed in the last hundred years, and we prefer that to many modern hotels, which tend to look the same as each other. The Wellington has character! Our room was very comfortable and quite spacious. We can strongly recommend the breakfast, though we had to wait for a table as the hotel was so full. That was a bit annoying, and there was also nowhere to sit in the lounge.

C) We made our reservation by phone without problem, but when we arrived the receptionist couldn't see it on the computer system. Luckily there was a room available. It wasn't quite what we would have chosen, but it was a pleasure to sit in it with a cup of tea, and look out at the swimmers and surfers in the sea.

D) We'd be happy to stay at the Wellington again. Although there's nothing special about the rooms, the view from the lounge is lovely, and the restaurant staff were friendly and efficient. Breakfast was a highlight – there was so much on offer we could hardly decide what to eat. We'd stay another time just for that!

E) The staff all did their jobs efficiently, and were very helpful when we asked for information about the area. The only difficulty we had was making our reservation online – it wasn't clear whether payment for our deposit went through or not, and I had to call the hotel to find out. Still, once we'd arrived, everything went very smoothly, and we had a delicious dinner in the restaurant.

PART 3

Questions 14–20

The reading passage has seven sections, A-G. Choose the correct heading for each section from the list of headings below. There are more headings than sections, so you will not use all of them. Write the correct number, i-x.

List of Headings:

i) A decrease in the zebra population
ii) An obstruction on the traditional route
iii) An unknown species
iv) Some confusing information
v) Staying permanently in the Makgadikgadi
vi) Nearly a record in the zebra world
vii) Three different ways of living
viii) The original aim of the work
ix) How was the information passed on?
x) Why it is important to study zebras

14. Paragraph A **15. Paragraph B** **16. Paragraph C**
17. Paragraph D **18. Paragraph E** **19. Paragraph F**
20. Paragraph G

THE ZEBRAS' LONG WALK ACROSS AFRICA

A. For any animal to travel over 270 km in Botswana partly across the sand and low bush terrain of the Kalahari Desert is a remarkable achievement. But to do so in 11 days and without any obvious motivation, as this zebra population does, is quite extraordinary. On average their journey involves an exhausting round-trip of 588 km – between the Makgadikgadi salt pan area and the Okavango river – making it second only to the great trek undertaken by the zebra herds in the Serengeti National Park. However, what is even more incredible still in my view is that until recently it was completely unheard of.

B. Hattie Bartlam, a researcher, discovered this migration while she was tracking zebra groups, officially known as harems, by the Okavango River for her PhD, Each harem consists of a stallion and his seven or eight mares with juvenile foals. There is no loyalty between zebras beyond this social group, though harems often gather together into so- called herds. For her study, Hattie had planned to compare the small-scale movement patterns of 11 different zebra herds in the area.

C. In December, when the annual rains had transformed the roads into rivers, Hattie was, therefore, more than a little surprised when she checked the data sent by the radio collars she fits to the zebras she is tracking to find that six of the harems were 270 km away on the edge of the Makgadikgadi, a huge mineral-rich area where salt has collected over the years as water evaporates in the heat. Then, when the last of the moisture from the rains had disappeared in May the following year, five of those harems came wearily back to the Okavango. This raised the question: why, despite a plentiful supply of food and water, were the zebras being drawn eastwards to the salt pans? Even more difficult to understand was what made six of the groups travel so far, while the other five remained by the Okavango.

D. This discovery created quite a buzz in the research community. I decided to visit Hattie and she explained that a century ago the large number of Botswana's zebra and wildebeest herds and the resulting competition for grass made migration essential. One of the migration tracks went from the Okavango to Makgadikgadi. But in the late 1960s, giant fences were put up to stop foot and mouth and other diseases spreading between wildlife and domestic cattle. One of these went across the migration track. Though the animals could get round the obstacle, each leg of their journey would now be 200 km longer – an impossible distance given the lack of permanent water on the extended route. Even today, with the fence gone (it was taken down in 2004), there is dangerously little drinking water to support the zebras on the return journey to the Okavango.

E. As a zebra can live up to 20 years, the migration must have skipped at least one generation during the 40 or so years that the fences were up. This prompts another question: it has always been assumed that the young of social herbivores like zebras learn migratory behaviour from their parents, so how did the latest generation learn when and where to go? Not from their parents, who were prevented from migrating. Did they follow another species, such as elephants? We may never know.

F. Hattie's data points to the conclusion that there are several zebra populations adopting different behaviour. The first, like the vast majority of the Okavango zebras, take it easy, spending the entire year by the river. The second group, 15,000-20,000 strong, work a bit harder. They divide their time between the Makgadikgadi salt pans and the Boteti River, which is reasonably nearby. They sometimes struggle to find water in the Boteti area during the dry season, often moving 30 km in search of fresh grazing. Their reward: the juicy grass around the Makgadikgadi after the rains. The final group of zebras, whose numbers are more modest (though as yet unknown), must surely be considered as among the animal kingdom's most remarkable athletes. By moving between the Okavango and the salt pans, they enjoy the best of both worlds. But the price they pay is an extraordinary journey across Botswana.

G. Endangered species naturally tend to grab the headlines, so it's refreshing for a relatively abundant animal like the zebra to be the centre of attention for once. Zebras are a vital part of the food chain: understanding their migration, in turn, helps us to interpret the movements of their predators, and Hattie's research has shed light on the impact of fences on migratory animals. So what triggered her interest in zebras? She explains that it is easier to get funding to study exciting animals like lions. Crucial as that undoubtedly is, she believes that herbivores like zebras are key to understanding any ecosystem. The scientific community is fortunate that people like Hattie are willing to take the hard option.

PART 4

Questions 21–29

Read the following text for questions 21-29.

HOW BACTERIA INVENTED GENE EDITING

This week the UK Human Fertilisation and Embryology Authority okayed a proposal to modify human embryos through gene editing. The research, which will be carried out at the Francis Crick Institute in London, should improve our understanding of human development. It will also undoubtedly attract controversy - particularly with claims that manipulating embryonic genomes is a first step towards designer babies. Those concerns shouldn't be ignored. After all, gene editing of the kind that will soon be undertaken at the Francis Crick Institute doesn't occur naturally in humans or other animals.

It is, however, a lot more common in nature than you might think, and it's been going on for a surprisingly long time - revelations that have challenged what biologists thought they knew about the way evolution works. We're talking here about one particular gene editing technique called CRISPR-Cas, or just CRISPR. It's relatively fast, cheap and easy to edit genes with CRISPR - factors that explain why the technique has exploded in popularity in the last few years. But CRISPR wasn't dreamed up from scratch in a laboratory. This gene editing tool actually evolved in single-celled microbes.

CRISPR went unnoticed by biologists for decades. It was only at the tail end of the 1980s that researchers studying Escherichia coli noticed that there were some odd repetitive sequences at the end of one of the bacterial genes. Later, these sequences would be named Clustered Regularly Interspaced Short Palindromic Repeats - CRISPRs. For several years the significance of these CRISPRs was a mystery, even when researchers noticed that they were always separated from one another by equally odd 'spacer' gene sequences.

Then, a little over a decade ago, scientists made an important discovery. Those 'spacer' sequences look odd because they aren't bacterial in origin. Many are actually snippets of DNA from viruses that are known to attack bacteria. In 2005, three research groups independently reached the same conclusion: CRISPR and its associated genetic sequences were acting as a bacterial immune system. In simple terms, this is how it works. A bacterial cell generates special proteins from genes associated with the CRISPR repeats (these are called CRISPR associated - Cas - proteins). If a virus invades the cell, these Cas proteins bind to the viral DNA and help cut out a chunk. Then, that chunk of viral DNA gets carried back to the bacterial cell's genome where it is inserted - becoming a spacer. From now on, the bacterial cell can use the spacer to recognise that particular virus and attack it more effectively.

These findings were a revelation. Geneticists quickly realised that the CRISPR system effectively involves microbes deliberately editing their own genomes - suggesting the system could form the basis of a brand new type of genetic engineering technology. They worked out the mechanics of the CRISPR system and got it working in their lab experiments. It was a breakthrough that paved the way for this week's announcement by the HFEA. Exactly who took the key steps to turn CRISPR into a useful genetic tool is, however, the subject of a huge controversy. Perhaps that's inevitable - credit for developing CRISPR gene editing will probably guarantee both scientific fame and financial wealth.

Beyond these very important practical applications, though, there's another CRISPR story. It's the account of how the discovery of CRISPR has influenced evolutionary biology. Sometimes overlooked is the fact that it wasn't just geneticists who were excited by CRISPR's discovery - so too were biologists. They realised CRISPR was evidence of a completely unexpected parallel between the way humans and bacteria fight infections. We've known for a long time that part of our immune system "learns" about the pathogens it has seen before so it can adapt and fight infections better in future. Vertebrate animals were thought to be the only organisms with such a sophisticated adaptive immune system. In light of the discovery of CRISPR, it seemed some bacteria had their own version. In fact, it turned out that lots of bacteria have their own version. At the last count, the CRISPR adaptive immune system was estimated to be present in about 40% of bacteria. Among the other major group of single-celled microbes - the archaea - CRISPR is even more common. It's seen in about 90% of them. If it's that common today, CRISPR must have a history stretching back over millions - possibly even billions - of years. "It's clearly been around for a while," says Darren Griffin at the University of Kent.

The animal adaptive immune system, then, isn't nearly as unique as we thought. And there's one feature of CRISPR that makes it arguably even better than our adaptive immune system: CRISPR is heritable. When we are infected by a pathogen, our adaptive immune system learns from the experience, making our next encounter with that pathogen less of an ordeal. This is why vaccination is so effective: it involves priming us with a weakened version of a pathogen to train our adaptive immune system. Your children, though, won't benefit from the wealth of experience locked away in your adaptive immune system. They have to experience an infection - or be vaccinated - first hand before they can learn to deal with a given pathogen.

CRISPR is different. When a microbe with CRISPR is attacked by a virus, the record of the encounter is hardwired into the microbe's DNA as a new spacer. This is then automatically passed on when the cell divides into daughter cells, which means those daughter cells know how to fight the virus even before they've seen it. We don't know for sure why the CRISPR adaptive immune system works in a way that seems, at least superficially, superior to ours. But perhaps our biological complexity is the problem, says Griffin. "In complex organisms any minor [genetic] changes cause profound effects on the organism," he says. Microbes might be sturdy enough to constantly edit their genomes during their lives and cope with the consequences - but animals probably aren't. The discovery of this heritable immune system was, however, a biologically astonishing one. It means that some microbes write their lifetime experiences of their environment into their genome and then pass the information to their offspring – and that is something that evolutionary biologists did not think happened.

Darwin's theory of evolution is based on the idea that natural selection acts on the naturally occurring random variation in a population. Some organisms are better adapted to the environment than others, and more likely to survive and reproduce, but this is largely because they just happened to be born that way. But before Darwin, other scientists had suggested different mechanisms through which evolution might work. One of the most famous ideas was proposed by a French scientist called Jean-Bapteste Lamarck. He thought organisms actually changed during their life, acquiring useful new adaptations non-randomly in response to their environmental experiences. They then passed on these changes to their offspring.

People often use giraffes to illustrate Lamarck's hypothesis. The idea is that even deep in prehistory, the giraffe's ancestor had a penchant for leaves at the top of trees. This early giraffe had a relatively short neck, but during its life it spent so much time stretching to reach leaves that its neck lengthened slightly. The crucial point, said Lamarck, was that this slightly longer neck was somehow inherited by the giraffe's offspring. These giraffes also stretched to reach high leaves during their lives, meaning their necks lengthened just a little bit more, and so on. Once Darwin's

ideas gained traction, Lamarck's ideas became deeply unpopular. But the CRISPR immune system - in which specific lifetime experiences of the environment are passed on to the next generation - is one of a tiny handful of natural phenomena that arguably obeys Lamarckian principles.

"The realisation that Lamarckian type of evolution does occur and is common enough, was as startling to biologists as it seems to a layperson," says Eugene Koonin at the National Institutes of Health in Bethesda, Maryland, who explored the idea with his colleagues in 2009, and does so again in a paper due to be published later this year. This isn't to say that all of Lamarck's thoughts on evolution are back in vogue. "Lamarck had additional ideas that were important to him, such as the inherent drive to perfection that to him was a key feature of evolution," says Koonin. No modern evolutionary biologist goes along with that idea. But the discovery of the CRISPR system still implies that evolution isn't purely the result of Darwinian random natural selection. It can sometimes involve elements of non-random Lamarckism too – a "continuum", as Koonin puts it. In other words, the CRISPR story has had a profound scientific impact far beyond the doors of the genetic engineering lab. It truly was a transformative discovery.

Choose the correct letter, A, B or C.

21 **The research carried out at the Francis Crick Institute in London is likely to be controversial.**
A) True
B) False
C) Not Given

22 **Gene editing, like the one in the upcoming research, can happen naturally in humans or other animals.**
A) True
B) False
C) Not Given

23 **CRISPR-Cas is a gene editing technique.**
A) True
B) False
C) Not Given

24 **CRISPR was noticed when the researchers saw some odd repetitive sequences at the ends of all bacterial genes.**
A) True
B) False
C) Not Given

25 **A group of American researchers made an important revelation about the CRISPR.**
A) True
B) False
C) Not Given

Choose the correct letter, A, B, C or D.

26 **'Spacer' sequences look odd because:**
A) they are a bacterial immune system
B) they are DNA from viruses
C) they aren't bacterial in origin
D) all of the above

27 **The ones, who were excited about the CRISPR's discovery, were:**
A) biologists
B) geneticists
C) physicists
D) A and B

28 **Word "learns" in the line 44, 6th paragraph means:**
A) determines
B) gains awareness
C) adapts
D) studies

29 **What makes CRISPR better than even our adaptive immune system?**
A) long history of existence
B) immortality
C) heritability
D) adaptiveness

PART 5 Questions 30–35

Read the following text for questions 30-35.

HOW TO DEAL WITH THE ANNUAL PERFORMANCE APPRAISAL

The annual performance appraisal can help improve your productivity and provide a foundation for your work priorities. It is, however, critical to have the right attitude and approach.

Knowing what areas your superiors see as your weaknesses is the most direct way of increasing the likelihood of being considered for promotion, if that is what you are looking for.

Preparation Send your boss a summary of your achievements. Reminding your boss of activities, special assignments you did, and projects you were in charge of helps him or her create a more accurate performance appraisal. Consider keeping notes of these on a regular basis to make it easier to provide the data when required.

Create a list of questions you would like to discuss during your appraisal. This one-on- one time with your boss is an excellent opportunity to ask him or her about your role in the company, request any additional responsibilities you would like and clarify your priorities. But it is best to focus your attention around personal and professional improvements, rather than financial considerations, such as an increase in salary.

During the appraisal Present a positive attitude as soon as you enter the appraisal room. This approach may lead to a more constructive discussion of review items. Avoid taking any negative assessments that are offered as a personal attack, but rather try to take them on board calmly, because if you put the failings right you will improve your performance. A realistic assessment of your strengths and weaknesses can be one of the most beneficial ways of helping you advance in the company.

After the appraisal Create a list of personal goals based on your performance appraisal. Make the items detailed and measurable if possible. Send this list to your boss so he or she knows you took the appraisal seriously. Use this list to help achieve higher scores on your next performance appraisal.

Six months after the appraisal, ask for a mid-term review with your boss to discuss your progress. This session should be more relaxed and informal than the official review. Ask for more

feedback to help you improve. Checking in with your boss helps him or her remember your dedication as far as your job is concerned, and may help remove any criticisms before they become a review point on your next formal appraisal.

Complete the sentences below. Choose ONE WORD ONLY from the text for each answer.

By learning at an appraisal what areas of work need improving, staff can improve their chances of getting **30** _____.

It is important to think of some **31** _____ that can be used during the appraisal.

The appraisal can be a good time to ask the boss for extra **32** _____.

React **33** _____ to any criticism.

It is helpful to identify a number of individual **34** _____ arising from the appraisal comments.

Staff can request a meeting half-way through the year to look at the **35** _____ which has been achieved.

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35 answers

1 ice
2 Arctic
3 sea
4 found
5 These / Polar
6 group
7 C
8 A
9 B
10 E
11 D
12 A
13 B
14 vi
15 viii
16 iv
17 ii
18 ix
19 vii
20 x
21 True
22 False
23 True
24 False
25 Not Given
26 C / they aren't bacterial in origin
27 D / A and B
28 B / gains awareness
29 C / heritability
30 promotion
31 questions
32 responsibilities
33 calmly
34 goals
35 progress